Preliminary Insights to enable automation of the Software Development Process in Software StartUps

An Investigation Study from the use of Artificial Intelligence and Machine Learning

Olimar Borges
PUCRS University
Porto Alegre, Brazil
olimar.borges@edu.pucrs.br

Valentina Lenarduzzi
University of Oulu
Oulu, Finland
valentina.lenarduzzi@oulu.fi

Rafael Prikladnicki
PUCRS University
Porto Alegre, Brazil
rafaelp@pucrs.br

ABSTRACT

Artificial Intelligence (AI) and Machine Learning (ML) tools and techniques have increasingly effectively supported Software Engineering (SE) tasks, whether for requirements classification, software refactoring, defect prediction, and many others. In the context of software StartUps, where innovative and scalable software products are developed, dealing with the pressure of fast delivery of a working solution becomes a challenging factor. We aim to investigate AI and ML techniques used by SE practitioners and entrepreneurs to support their Software Development Processes (SDP) and thus enable their use by software StartUps. We seek to identify this information through the application of an online Survey instrument, mainly disseminated in Brazil and Finland. This preliminary study provides insights that can support improving the SDP in StartUps.

KEYWORDS

Development Process, Artificial Intelligence, Machine Learning, StartUp, SWEBOK

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1 INTRODUCTION

Undoubtedly, the software is increasingly integrated into our routine and lifestyle, which emphasizes the care to improve the quality of the software as a major challenge for software engineering. However, research at the intersection of SE and AI is emerging as an important means of tackling the big challenges ahead [7].

In intelligent software engineering to address these challenges, AI offers various technologies for reasoning, problem-solving, planning, learning, and more, like ML. Identifying the best practices for SDP automation can bring significant benefits to practitioners and entrepreneurs inside software StartUps. Since the engineering context in StartUps is characterized by uncertainty, lack of resources, rapid evolution, and an immature team, among other factors [6]. The solution of SE problems brings significant benefits to the software industry. Therefore, several algorithms and techniques have been proposed and improved over time to make prediction systems effectively applied in practice [4].

We conducted a Survey with the aim to gather experiences and insights on how AI and ML are used in the SDP of the software StartUp projects. The main objective of a Survey is to collect information from a specific group of people, through a sample of individuals, from a large population [5]. Although the results are preliminary, we identified possible directions needed to improve research in the area and possibilities for future work.

2 THE EXPLORATORY STUDY

The *goal* of this study is improve SDP, composed of AI/ML components, to support and automate the development process in software StartUps. Based on our goal, we defined the following research question: **RQ**. *How is Artificial Intelligence/Machine Learning used in the Software Development Process?*

We developed a Survey prototype to ascertain the feasibility study of the implementation of ML in the industry.We designed the pilot study to outline profile patterns of groups of participants, companies, and teams using techniques and tools for the use of ML in the SDP of software StartUps. We divided the Survey into seven groups with 26 questions, of which 17 are relevant to the survey questions in this article. The other 8 questions are about respondent demographic information, and 1 question asks for general feedback regarding the survey instrument. We framed the analyses following the concepts of the first ten Knowledge Areas (KA) of SWEBOK [3] as a guide to structure the results (**KA1:** Software Requirements; **KA2:** Software Design; **KA3:** Software Construction; **KA4:** Software Testing; **KA5:** Software Maintenance; **KA6:** Software Configuration Management; **KA7:** Software Engineering Management; **KA8:** Software Engineering Process; **KA9:** Models and Methods of Software Engineering; **KA10:** Software Quality). We designed this version based on the results obtained in the study conducted by Borges et al. [1] and the complete instrument is available in our replication package[1].

3 PRELIMINARY RESULTS AND DISCUSSION

We summarize our interpretation and discussion of the results divided by the survey question groups, as follows:

[1] https://figshare.com/s/b1a9eca4b7fd2a603cde

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SWEBOK: The KA1, KA2, and KA3 are little explored by the use of AI/ML, even though there are still a few studies that explore them. Such KA's are more dependent on human perceptions and more difficult to fully automate. Studies usually present semi-automatic ways to help in these phases, but human intervention ends up being necessary. On the other hand, KA4, KA5, KA7, KA8, K9, and KA10 are frequently cited. While KA4 and KA10 are the most discussed in literature studies, as they are possible phases of complete automation and are often approached together. And KA9 is directly related to automation, as it focuses on the direct use and creation of models, methods, tools, algorithms and other approaches for SDP. There are several studies that propose the creation of these approaches to help a certain stage of development.

Techniques and Tools: We initially asked which tools the respondents had already used based on previous studies [1]. Among the tools, WEKA, MATLAB, and Scikit-learn are the most popular, followed by libsvm, Mulan, RNNLM Toolkit, TensorFlow, R language and tools based in Python in general. Among the responses obtained, was reported that frequently uses AI/ML for SE tasks in all KA's. In the same time, some of the respondents said that never used it for KA1, KA2, and KA3, using it a few times for KA4 and KA6 but used it frequently for KA5, KA7, KA8, KA9, and KA10. When asked about AI/ML algorithms and techniques, most of the algorithms mentioned usually belong to Supervised Learning for classification (such as NB, RF, DT, J48, KNN, and SVM) or regression (such as LR and SVR). Or even, for both, as Neural Networks (ANN, MLP, CNN, etc.), not limited to them, but also for Learning Sets (such as Ba, AB, Bo). As for Unsupervised Learning, those mentioned are usually for clustering, such as K-means.

Product, Feature or Service: For the analyzed answers, the respondents directed their creations mainly to KA5, as the descriptions about the creation of their products and services help developers in the software maintenance phases, whether in the construction and maintenance of research tools, as for the collection and analysis of documents by score.

Priority: One of the primary purposes of asking which KA's respondents would like to automate, in order of priority, is to understand which KA's, from their perspective, are of most interest for automation. The interpreted results help us understand the greater interest in certain KA's concerning others. The project's purpose, the time for its creation, and other factors related to project management are not always favorable to focusing on a KA. In this way, the attempt for automation can become unfeasible in the context it is inserted. Or, the lack of broader knowledge about a specific KA can be an aversion factor in the search for automation methods.

Challenges: We can find several related challenges in the literature regarding AI/ML to support SE [2]. In this sense, it was mentioned that for KA5, data's low quality was the main challenge for *automation* and the need to clean and process this data as a solution. We verified that this perception and the literature converge concerning this challenge in automation, opening another phase opportunity with great interest in exploring its automation. Compared to previous analyses, even though KA5 appears more prone to automation, there are still challenges to be addressed. Regarding the challenges to solve SE tasks, the KA's listed (KA3, KA4, and KA5) are well related, as they are areas usually worked together. Therefore, there are challenges that intersect between them, including their integration, which was mentioned by one of the respondents. While you looked at the KA's with the most solutions found to solve, KA1 and KA8 were the most mentioned, suggesting that respondents find more solutions to tasks related to these areas.

Closure: We asked what would be the *dreams tool* that would help them to automate the SDP using AI/ML. In the analysis, four respondents described a tool more targeted to the KA3 phase.Analyzing the answers, we found that the suggestions are tools aimed at the software construction phase, which presents another exciting opportunity to explore automation. KA4 is one of the phases with more works in the literature to solve problems related to SE tasks. Noting that the respondents still need tools that support it, it is an exciting point to verify that even so, the area still has a lot of space to be explored. As a goal, this KA focuses on planning and controlling the software development environment. The software architecture is important because it defines the basis of software. We highlight that this KA is another essential addition to our analysis.

4 CONCLUSION REMARKS

We proposed a preliminary investigation of the use of AI and ML by entrepreneurs and practitioners in their product and service development processes. From the analysis, we identified possible directions needed to improve the automation of some of the KAs. We found that KA7 and KA8 are more interested in automation, as the resolution of their tasks helps the management process for software managers. We verified that there is a great interest in solutions within KA5 since they can speed up the maintenance of the software, but that the low quality of the data is still a challenge in this process. The KA2 was listed as the highest priority for automation since it consists of the base structure of the software, and any faulty construction can result in significant problems during the project. Even though many studies focus on tasks linked to KA4, there is still interest and space to be explored for solutions that support processes in this area. This study is part of a set of researches carried out at the intersection of SE and AI for use by software StartUps. We hope that the preliminary results of this study guide software StartUps in its development processes, minimally enabling the use of AI/ML during all phases of its software development.

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